require the same. We prefer to use mole concept for non-redox reactions and gram equivalent concept for redox reactions.

For example,

If we know the equivalent mass of $KMnO_4$ and anhydrous ferrous sulphate, without writing balanced chemical reaction we can straightaway say that 31.6 g of $KMnO_4$ reacts with 152 g of $FeSO_4$ using gram equivalent concept.

The same can also be explained on the basis of mole concept. The balanced chemical equation for the above mentioned reaction is

$$10\ FeSO_4 + 2\ KMnO_4 + 8\ H_2SO_4 \rightarrow K_2SO_4 + 2\ MnSO_4 + 5\ Fe_2(SO_4)_3 + 8H_2O$$

i.e. 2 moles ($2 \times 158 = 316$ g) of potassium permanganate reacts with 10 moles ($10 \times 152 = 1520$ g) of anhydrous ferrous sulphate.

$\therefore$ 31.6 g $KMnO_4$ reacts with

$$\frac{1520}{316} \times 31.6 = 152 \text{ g of } FeSO_4$$

Evaluate Yourself

4a) 0.456 g of a metal gives 0.606 g of its chloride. Calculate the equivalent mass of the metal.

4b) Calculate the equivalent mass of potassium dichromate. The reduction half-reaction in acid medium is,

$$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2\ Cr^{3+} + 7H_2O$$

1.6 Empirical Formula and Molecular Formula

Elemental analysis of a compound gives the mass percentage of atoms present in the compound. Using the mass percentage, we can determine the empirical formula of the compound. Molecular formula of the compound can be arrived at from the empirical formula using the molar mass of the compound.

Empirical formula of a compound is the formula written with the simplest ratio of the number of different atoms present in one molecule of the compound as subscript to the atomic symbol. Molecular formula of a compound is the formula written with the actual number of different atoms present in one molecule as a subscript to the atomic symbol.

Let us understand the empirical formula by considering acetic acid as an example.

The molecular formula of acetic acid is $C_2H_4O_2$

The ratio of C : H : O is 1 : 2 : 1 and hence the empirical formula is CH_2O.

1.6.1 Determination of Empirical Formula from Elemental Analysis Data :

Step 1: Since the composition is expressed in percentage, we can consider the total mass of the compound as 100 g and the percentage values of individual elements as mass in grams.

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